

Research Interests

Humanoid Robot Learning; Generalizable Physical Skill Learning; Loco-Manipulation; Reinforcement Learning; Motion Priors and Imitation Learning; Perception for Physical Control; Sim-to-Real and Learned Dynamics; Contact-Rich Robot Control

Education

Nanjing University of Posts and Telecommunications (NJUPT)

B.Eng. Candidate in Artificial Intelligence | Expected Graduation: 2028

GPA: 4.08/5.0 | Ranked 3/133 in Artificial Intelligence

Research Experience

Learning Dynamic Skills for Humanoid Robots via Reinforcement Learning

2026–Present

T1 Humanoid Kicking, Locomotion, and Skill Generalization

- Implemented and modified reinforcement-learning environments for humanoid locomotion and contact-rich kicking tasks using MuJoCo/MJLab/Warp simulation workflows.
- Implemented and evaluated task components including action and observation spaces, ball reset logic, contact detection, reward functions, termination conditions, and curriculum learning procedures.
- Conducted experiments to analyze policy behavior through contact quality, ball velocity, target direction error, post-kick recovery, and training-stability metrics.
- Investigating Conditional AMP and motion-prior-based policies for skill generalization across different ball positions, kicking directions, and task conditions.

Conditional AMP for Generalizable Humanoid Kicking

2026

RL Research Project

- Investigated distance-conditioned Conditional AMP for humanoid kicking using condition vectors over target distance and kick type with curriculum learning.
- Studying how expert-policy condition alignment and motion priors influence training stability and cross-condition skill generalization.

Motion Prior Learning for Generalizable Humanoid Skills

2026

Research Exploration

- Studying AMP-style motion priors, human motion data, encoder-decoder policies, and latent representations for generalizable humanoid skill learning.
- Exploring how motion priors can support natural, stable, and transferable behaviors without sacrificing task completion.

Robot Hand-Eye Calibration with 3D Sensing and Geometric Constraints

2026

Research Project | Manuscript in Preparation | Nanjing University of Posts and Telecommunications

- Conducting research on robot hand-eye calibration with a completed patent application and ongoing manuscript preparation.
- Investigating multi-stage calibration methods based on orthogonal geometric constraints, point-cloud filtering, and optimization.
- Conducted calibration experiments with ABB industrial robots and 3D sensing systems for evaluation.

Technical Skills

Programming: Python, C++, Git/GitHub, Linux

Machine Learning / RL: PyTorch, Reinforcement Learning, PPO, reward design, curriculum learning

Robot Learning: Generalizable skill learning, humanoid motion control, loco-manipulation concepts, motion imitation, AMP-style motion priors

Simulation: MuJoCo, MJLab, Warp

Robotics Software: ABB RAPID Programming, Robot Calibration, 3D Sensing, Point-Cloud Processing

Awards

- Third Place, RoboCup China Open 3D Simulation League, 2025.
- Third Place, Jiangsu Provincial Robot Competition 3D Simulation Group, 2025.
- Third Prize, National Undergraduate Electronic Design Contest, Jiangsu Province, 2024.